Signal Models for Changes in Polarimetric SAR Data

Armando Marino, *Member, IEEE*, and Matteo Nannini

***Abstract*— Synthetic aperture radar (SAR) polarimetry can improve change detection in terms of detection capabilities. In this work, we are proposing to extend the idea of target decomposition to changes affecting partial targets. This will allow the separation of polarimetric-dependent changes, providing extra information that can be used to better understand the processes affecting the targets. Three models for changes are proposed and compared. The methodologies are based on Lagrangian optimizations of distinct operators built using quadratic forms for a power ratio and a power difference. The optimizations can be accomplished by diagonalizations of specific matrices derived from polarimetric covariance matrices. These are, therefore, spectral decompositions of an appropriate matrix which we define as change matrix. The theoretical validity of the models is assessed using Monte Carlo simulations. Additionally, we perform real data validation exploiting L-band quad-polarimetric data from the E-SAR (DLR) SARTOM 2006 campaign and ALOS PALSAR (JAXA) acquisitions in Morecombe Bay (U.K.). We observed that the two algorithms based on power difference allow to decompose the change into the minimal set of scattering mechanisms (SMs) that have been added or removed from the scene. The two algorithms differ on the initial assumption on the change. On the other hand, the ratio operator provides a better detection performance although the eigenvalues do not correspond to meaningful SMs. A combination of the three methodologies can, therefore, improve detection and classification of changes.**

***Index Terms*— Change detection, decompositions, polarimetry, signal models, synthetic aperture radar (SAR).**

I. Introduction

CHANGE detection is a valuable topic in synthetic aperture radar (SAR) remote sensing. Polarimetric SAR (PolSAR) can improve the results of single polarimetric algorithms [1]–[10]. The polarization of the transmitted and received waves can be exploited to extract more information about the observed targets. This information enhances classification. In this work, we want to propose decompositions to help classify the change.

It is always possible to perform approaches where the two set of PolSAR acquisitions are analyzed separately. However, due to the complexity of the multidimensional SAR data, it is generally hard and not straightforward to identify the scattering mechanisms (SMs) which have suffered the change. Specifically, from an algebraic point of view, there is no methodology to identify the minimal unique set of SMs that have changed. Often, we can just measure the increase or decrease of a set of targets (surface, dihedral [11], [12]) or parameters (entropy, α [13]), but not point at the exact SM that produced such changes. In this work, we want to propose an operation that is able to synthesize the change and reveal in a straightforward way the SM which produced the maximum change.

In the following, there is a brief introduction to PolSAR tools that we will use. A single target is defined as a deterministic target that does not change its polarimetric behavior in time/space. Therefore, it can be represented by a single scattering matrix $[S]$ or equivalently a single scattering vector [14], [15]:

$$\underline{k}_L = [\mathrm{HH},\ \mathrm{HV},\ \mathrm{VH},\ \mathrm{VV}]^{\mathrm{T}} \tag{1}$$

where H and V stands for linear horizontal and vertical and the repeated letter indicates the transmitter–receiver sequence. The previous is obtained employing the Lexicographic basis set. In the case of a reciprocal medium and monostatic sensor, HV = VH and $\underline{k}$ is 3-D complex (i.e., $\underline{k}_P \in \mathbb{C}^3$) [14]. Another largely used basis set to convert $[S]$ into a scattering vector is the Pauli basis. In the reciprocal case, this is $\underline{k}_P = 1/\sqrt{(2)}[\mathrm{HH}+\mathrm{VV}, \mathrm{HH}-\mathrm{VV}, 2*\mathrm{HV}]^{\mathrm{T}}$. An SM $\underline{\omega}$ is an ideal target and is defined as a normalized scattering vector.

The targets observed by an SAR system are often distributed over an area larger than the resolution cell and composed by different objects. For this reason, each pixel of the image containing such distributed targets shows a specific polarimetric behavior. Such targets take the name of partial targets and they are generally described exploiting the second-order statistics [14], [15]. In this context, a target covariance matrix can be estimated as $[C] = \langle \underline{k}\ \underline{k}^{*T} \rangle$, where $\langle . \rangle$ is the finite averaging operator. In case that the single look complex (SLC) pixel can be modeled by a Complex Gaussian, the second-order statistics are necessary and sufficient to completely characterize a partial target (and the covariance matrix is Wishart distributed). In case of the Pauli basis, the covariance matrix is often named coherency matrix.

An important concept exploited in this work is the quadratic form of a covariance matrix

$$P_T = \underline{\omega}^{*T}[C]\underline{\omega}. \tag{2}$$

The latter represents the power backscattered by the SM represented by ω. This is provided the SM is present.

Manuscript received January 15, 2021; revised June 14, 2021 and August 6, 2021; accepted August 30, 2021. Date of publication October 6, 2021; date of current version January 26, 2022. *(Corresponding author: Armando Marino.)*

Armando Marino is with the Faculty of Natural Sciences, University of Stirling, Stirling FK9 4LA, U.K. (e-mail: armando.marino@stir.ac.uk).

Matteo Nannini is with the Microwaves and Radar Institute, German Aerospace Center (DLR), 82234 Wessling, Germany.

Digital Object Identifier 10.1109/TGRS.2021.3113182

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

Otherwise, it just represents the power on that projection vector ω.

The novelties of this article are the following: 1) developing and validating for the first time (in a peer reviewed publication) the change detector based on optimization of DIFFERENCE of power [16] and 2) provide a comparison using simulated and real data of three methodologies based on optimizations of quadratic forms for change detection including [17]–[19].

II. Signal Models for Change Detection With Polarimetric Data

The starting point of this work is the consideration that the power of an SM can be estimated using a quadratic form of the covariance matrix $[C]$ and the appropriate projection vector $\underline{\omega}$. In the following, we will present analyses using a general covariance matrix; however, it is important to keep in mind that if we want to interpret the results in sight of the Cloude–Pottier decomposition [20], the scattering vector needs to be in Pauli basis.

The analyses in this article are based on the following two key assumptions.

1) We refer to "polarimetric SM." These are idealized targets that can be represented by a single and unique normalized scattering matrix (these can be different from interferometric SMs). We will also omit the word "polarimetric" unless this can produce confusion.
2) If there is a change of SM between two acquisitions, this will reveal itself in a change of the matrix $[C]$. Unfortunately, PolSAR signal decomposition is ill-posed and we can only monitor three independent targets (unless a physical model is used). If changes in the scene are such that the covariance matrix remains identical, then it is not possible to detect such changes using PolSAR.

The three change models proposed and compared here are based on spectral signal decompositions of what we define as change matrices. Therefore, the output of these three algorithms is sets of eigenvalues and eigenvectors. It is important to keep in mind that a change matrix does not necessarily represent a unique partial target, but a set of targets or transformations. In the following, we sometime refer to these optimizations as "change decompositions," however these are more similar to signal decompositions and therefore should not be confused with traditional "target decompositions" such as [11]–[13]. These concepts will be described in more detail and validated in the remainder of this article.

The first step in interpreting the new decompositions is to build a change framework based on detection hypotheses [21].

1) The null hypothesis H_0 states that there is no modification of SM.
2) The alternative hypothesis H_1 states that there is a change of SM.

Since we are interested in identifying optimal basis for change matrices, we can apply optimizations techniques

$$\underline{\omega}^r_{\max} = \underset{\underline{\omega}\in\Omega}{\text{Argmax}}\left[F([C_1],[C_2],\underline{\omega})\right] \tag{3}$$

where $F[.]$ is a function of covariance matrices for the two acquisitions ($[C_1]$ and $[C_2]$) and the projection vector $\underline{\omega}$. Ω is a unitary complex sphere, and therefore $\underline{\omega}\in\mathbb{C}^3$, $||\underline{\omega}|| = 1$.

Although many different functions can be designed, in the following, we focus on two main expressions for $F[.]$, since these are derived from the two most straightforward operations in change detection: the ratio and the difference.

Another final note is on the fact that all the detectors tested in this article are not scale-invariant. This is because, in general applications, the overall intensity (or SPAN) of the acquisitions provides useful information. Therefore, changes in span will affect the detectors. However, if we want to use these detectors in applications where the overall intensity is introducing errors (e.g., if there are calibration errors), then each of the covariance matrices could be normalized by their Trace before using the matrices in the detectors.

A. Multiplicative Model

A change detector based on the optimization of the power ratio was already proposed by the authors for PolSAR data [22]–[24]

$$F\left(\left[C_1,C_2,\underline{\omega}\right]\right) = M = \rho_{21} = \frac{\underline{\omega}^{*T}C_2\underline{\omega}}{\underline{\omega}^{*T}C_1\underline{\omega}} \tag{4}$$

$$\underline{\omega}^r_{\max} = \underset{\underline{\omega}\in\mathbb{C}^3}{\text{Argmax}}\left[\frac{\underline{\omega}^{*T}C_2\underline{\omega}}{\underline{\omega}^{*T}C_1\underline{\omega}}\right]. \tag{5}$$

Additionally, the optimization of this functional is not new to the community, since it was previously proposed few decades ago by Swartz *et al.* [17] and Novak *et al.* [25] with the polarimetric match filter (PMF). The PMF aims at optimizing the contrast between a target $[C_2]$ and clutter $[C_1]$ by selecting the projection vector that maximizes the ratio of the quadratic forms. It also has to be said that this optimization was previously known to the pattern recognition community and it sits in the context of linear discriminant analysis [26].

In our approach, the optimization can be implemented using a constrained Lagrangian methodology which returns a diagonalization

$$\begin{aligned} L &= \underline{\omega}^{*T}[C_2]\underline{\omega} - \lambda\left(\underline{\omega}^{*T}[C_1]\underline{\omega} - C\right) \\ \frac{\partial L}{\partial \underline{\omega}^{*T}} &= [C_2]\underline{\omega} - \lambda[C_1]\underline{\omega} = 0 \\ [C_1]^{-1}[C_2]\underline{\omega} &= \lambda\underline{\omega}. \end{aligned} \tag{6}$$

To conclude, $[C_1]^{-1}[C_2] = [U_r]^{*T}[\Sigma_r][U_r]$, where $[\Sigma_r] = \text{diag}(\lambda_{1r},\lambda_{2r},\lambda_{3r})$ with $\lambda_{1r},\lambda_{2r},\lambda_{3r}\in\mathbb{R}^+$ and the columns of $[U_r]$ are the eigenvectors. The maximum eigenvalue λ_{1r} represents the value of the maximum ratio. Interestingly, optimizing the ratio

$$\rho_{12} = \frac{\underline{\omega}^{*T}[C_1]\underline{\omega}}{\underline{\omega}^{*T}[C_2]\underline{\omega}} \tag{7}$$

ends up with the diagonalization of the matrix: $[C_2]^{-1}[C_1]$.

Therefore, the projection vector that suffers the maximum change $\underline{\omega}_{\max}$ is

$$\underline{\omega}_{\max} = \text{Argmax}\left[\underset{\underline{\omega}\in\Omega}{\text{Opt}}\rho_{12},\quad \underset{\underline{\omega}\in\Omega}{\text{Opt}}\rho_{21}\right]. \tag{8}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

It is interesting to note that $([C_1]^{-1}[C_2])^{-1} = [C_2]^{-1}[C_1]$, and therefore they will have the same eigenvectors, but inverted eigenvalues (elevating a matrix does not change the eigenvectors) [27]. We can define the change matrix to diagonalize as $[C_m] = [C_1]^{-1}[C_2]$. In [22], we showed how it can be proved that $[C_m]$ can be diagonalized and it has real positive eigenvalues. Additionally, the matrix is not Normal and therefore its eigenvectors are not bound to be orthogonal

$$\underline{\omega}_{\max} = \text{Argmax}\left[\max_{\underline{\omega}\in\Omega}(\rho_{12}), 1/\min_{\underline{\omega}\in\Omega}(\rho_{12})\right]. \tag{9}$$

The eigenvalues are included in the interval between zero and infinity. A change reducing power will appear close to zero, while a change increasing power will appear as a large number. With the goal of improving the visualization of the results, the eigenvalues can be inverted when they are smaller than 1. In other words,

$$\begin{cases} \hat{\rho}_{\max} = \rho_{\max}, & \text{if } \rho_{\max} \geq 1 \\ \hat{\rho}_{\max} = \dfrac{1}{\rho_{\max}}, & \text{if } \rho_{\max} < 1. \end{cases} \tag{10}$$

In the following, for the sake of brevity, we will refer to this change decomposition as RATIO.

B. Additive Model

Another model considers the difference of quadratic forms for the function $F[]$. Preliminary results were already presented at conferences [16], [28], [29] and recently a modification of this optimization was proposed in [19].

1) Difference Matrix: The function we want to optimize now is

$$F\left(\left[C_1, C_2, \underline{\omega}\right]\right) = D = \underline{\omega}^{*T} C_2 \underline{\omega} - \underline{\omega}^{*T} C_1 \underline{\omega}. \tag{11}$$

D can be rewritten as $D = \underline{\omega}^{*T}[C_2 - C_1]\underline{\omega}$. We can define the change matrix C_c as $C_c = C_2 - C_1$. C_c is a Normal matrix, since it is the difference of two Normal matrices. This means that the diagonal elements of C_c are real and the upper triangular part is the complex conjugate of the lower triangular part. However, there is a difference with ordinary coherency matrices which are also Hermitian. C_c is not bound to be positive semi-definite, because the diagonal element can be negative. This means that its quadratic forms (using a generic $\underline{\omega}$) or its Trace can be negative. In other words, $P = \underline{\omega}^{*T}[C_c]\underline{\omega} \in \mathbf{R}$ can be negative for some $\underline{\omega}$ and $\text{Trace}([C_c]) \in \mathbf{R}$ can be negative.

Not being bound to be positive semi-definite could be a desirable feature for the change matrix since it allows to retrieve if the change in the partial target is increasing or reducing the power of an SM.

To find the maximum and minimum $\underline{\omega}$, we can apply again the Lagrangian optimization for the quadratic form $\underline{\omega}^{*T}[C_c]\underline{\omega}$. That is

$$\underline{\omega}_{\max} = \underset{\underline{\omega}\in\Omega}{\text{Argmax}}\left[\underline{\omega}^{*T}[C_c]\underline{\omega}\right]. \tag{12}$$

By constraining $\underline{\omega}$ to be unitary, we can obtain the Lagrangian as

$$L = \underline{\omega}^{*T}[C_c]\underline{\omega} - \lambda\left(\underline{\omega}^{*T}\underline{\omega} + C\right). \tag{13}$$

After differentiating over $d\underline{\omega}^{*T}$ and setting the derivative equal to zero, we obtain the equation

$$C_c\underline{\omega} = \lambda\underline{\omega}. \tag{14}$$

The Lagrangian multipliers can be calculated by performing a diagonalization of the matrix $[C_c]$. Since $[C_c]$ is Normal, the eigenvalues will exist and be real, but not necessarily positive. This is because a change can increase or decrease the resulting power of an SM. Additionally, the eigenvectors are orthogonal, which allow this decomposition to capture the minimal change (please see Section IV-A for more details).

We will refer to this decomposition as DIFFERENCE, or DIFF in short.

2) Constrained Difference Matrix: In [19], a constraint of the difference matrix is proposed. The rational is to allow the final change matrix to regain the property of being positive semi-definite. In this way, the matrix will represent a single physically feasible partial target. Clearly, one covariance matrix can only represent one partial target and therefore we need to define in which time direction the change has happened. In other words, we need to define if one partial target has been added to or removed from the scene.

The optimization is again based on a diagonalization of a matrix $[C_p]$, where

$$\begin{cases} \left[C_p\right] = [C_2] - r_p[C_1]: & \text{if } r_p > r_m \\ \left[C_p\right] = [C_1] - r_m[C_2]: & \text{if } r_p < r_m \end{cases} \tag{15}$$

where

$$\begin{cases} r_p = \max \text{ eig}\left([C_1]^{-1}[C_2]\right) \\ r_m = \dfrac{1}{\max \text{ eig}\left([C_1]^{-1}[C_2]\right)}. \end{cases} \tag{16}$$

The optimization initially checks for the bigger r_p and r_m value and then produces a covariance matrix representing a partial target that has been either added or removed. The constraint ensures that the resulting matrix is positive semi-definite [19].

We will refer to this decomposition as PARTIAL DIFFERENCE, or ParDIFF in short. The word "partial" wants to lead the attention on the fact that the constraint produces a change matrix that can represent a partial target (if the assumptions for the change are met).

III. Theoretical Evaluation of Signal Models

The three previous decompositions are focused at very different typologies of changes, and therefore the eigenvectors have very different interpretations. To interpret these, we need to analyze the assumptions of the different physical models.

A. Additive Models: Unconstrained Optimization

In this model, the change is caused by an increase or reduction of power for an SM. The detection hypotheses for the addition case are in the following (the subtraction can be achieved by flipping the role of 1 and 2)

$$\begin{cases} H_0 : [C_2] = [C_1] \\ H_1 : [C_2] = [C_1] + [C_c]. \end{cases} \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

The Null hypothesis sees no change, while the Alternative hypothesis converts the first acquisition into the second by adding the partial targets enclosed in $[C_c]$.

We name this signal model as Additive, since the change is additive. This signal model covers all the circumstances where a target has been added or removed from the scene. For instance, in the agricultural context, we may have that the first scene only has scattering from the ground $[C_{1g}]$, while the second may have only horizontal dihedrals $[C_{2d}]$ due to plants. H_1 is true, with matrices $[C_c] = [C_{2d}] - [C_{1g}]$. The matrix $[C_c]$ will encapsulate that a dihedral SM was added and a surface SM was removed.

This decomposition is data-driven and when the number of SMs increases and surpasses the number of observables (the three polarization channels), the analysis becomes more complex and extra physical models need to be used to interpret the targets encapsulated in $[C_c]$ (as it is for the model-based decompositions that can have more than three components) [14].

The decomposition DIFF tries to provide an answer to the Additive hypothesis testing.

In the circumstance that the additive model is a good fit to reality, these results can be physically interpreted. The largest eigenvalue (if positive) represents the power of the largest SM that has been added to the scene, while the minimum (if negative) is the power of the SM that has been maximally removed from the scene. The maximum eigenvector represents the SM that has been added. The minimum eigenvector represents the SM that has been removed.

An interesting point is that the set of eigenvectors is orthogonal, because the matrix $[C_c]$ is Normal. This means that the eigenvectors also represent the minimal change that one could apply additively to transform the first partial target into the second partial target. It is minimal because the added and subtracted components are uncorrelated and therefore they have no projection on one another. If they were not orthogonal, one could always add the same partial target to both $[C_a]$ and $[C_s]$, and the result would not change (this would also have made the decomposition not unique). Orthogonality, therefore, is an important property for this decomposition.

It is also important to stress that the eigenvectors provide complementary information from just knowing the covariance matrices before $[C_1]$ and after $[C_2]$. It is decomposing the change into its orthogonal (and therefore minimal and unique) components.

This theoretical result will be tested in the following with Monte Carlo simulations and real data.

B. Additive Models: Constrained Optimization

In this model, the change is caused by either an increase or reduction of power of a partial target. The detection hypotheses for the addition case are

$$\begin{cases} H_0 : [C_2] = [C_1] \\ H_1 : [C_2] = r[C_1] + \left[C_p\right]. \end{cases} \tag{18}$$

The Null hypothesis sees no change, while the Alternative hypothesis converts a scaled first acquisition into the second by adding one partial target represented by $[C_p]$.

The change is additive although it is constrained by a scaling of the covariance matrix of the acquisition that is reducing. This signal model covers the circumstances where a unique partial target has been added or removed from the scene. For instance, in the agricultural context, we may have that the first scene only has ground component $[C_{1g}]$, while the second may have a combination of ground and horizontal dihedrals $r[C_{1g}] + [C_{2d}]$ due to plants. The scaling is needed to make sure that the change matrix $[C_p]$ can be interpreted as a unique feasible partial target.

$[C_p]$ is forced to be rank deficient (rank 2) by the constraint. It therefore only represents the two main spectral components of the added or removed target [19].

Since it is focused on a unique $[C_p]$ matrix, this decomposition is only valid when one unique partial target is either added or removed. The decomposition is resilient in dealing with situations when we have changes in both directions (part of a target is removed and a new target added), because r can be small to allow for reductions of one of the partial targets. However, if a partial target is completely removed $[C_1]$, the constraint will result in a very small r value. In fact, it can be proved that the only way to avoid negative eigenvalues for $[C_p]$ when we remove $[C_1]$ orthogonal to $[C_2]$ is by having $r_p = r_m = 0$. In general, the constraint will result in a $[C_p]$ that converges toward $[C_1]$ or $[C_2]$ when the assumption of a unique change is not met.

One may think to run two separate optimizations and obtain two matrices $[C_{p1}]$ and $[C_{p2}]$. This will not ensure that one of the resulting matrices may end up not being positive semi-definite (it was discouraged in [19]). Additionally, this will not remove the convergence effect. However, if we only focus on the dominant eigenvector of the two optimizations, the algorithm should still obtain estimations of the dominant change, providing the SMs that suffer the maximum addition or subtraction.

To conclude, in the circumstance that the additive model is a good fit and a unique partial target is added or removed, the decomposition assumptions are met. The largest eigenvalue represents the power of the largest SM that has been either added or removed. The maximum eigenvector represents the SM that has been added or removed.

This theoretical result is tested and validated in the following with Monte Carlo simulations and real data.

C. Multiplicative Model

In this model, we consider that a change in the partial target is not due to adding or removing SMs, but rather a transformation of the partial target itself. The detection hypotheses are

$$\begin{cases} H_0 : [C_2] = [C_1] \\ H_1 : [C_2] = [A][C_1]. \end{cases} \tag{19}$$

The change that transforms the second partial target into the first partial target is modeled as a multiplication by a transformation matrix $[A]$. We can derive that

$$[A] = [C_2][C_1]^{-1}. \tag{20}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

We can see that

$$[A]^{-1} = \left([C_2][C_1]^{-1}\right)^{-1} = [C_1]^{-1}[C_2] = [C_m]. \quad (21)$$

The matrix $[C_1]^{-1}[C_2] = [C_m]$ is the change matrix used in the spectral decomposition. This means that the eigenvectors of $[A]$ are the same as $[C_m]$ and they represent transformations.

When the multiplicative model is a good fit to reality, these results can be physically interpreted. The maximum eigenvector represents the direction in which the ellipsoid of the first partial target has been maximally stretched out to transform into the ellipsoid of the second partial target. On the other hand, the minimum eigenvector is the direction in which the ellipsoid has been squashed (compressed) to become the second ellipsoid. The eigenvalues present the ratio between the ellipsoid surface (i.e., the power) in the directions where the stretching happened.

Note that it is not possible to interpret these eigenvectors as SMs since they are directions of stretching for the partial targets. It is, therefore, more similar to a signal decomposition than a classic target decomposition and the optimal $\underline{\omega}$ are not SM.

This theoretical result will be tested in the following using Monte Carlo simulations and real data.

D. When to Use the Different Decompositions

1) *Additive unconstrained:* If partial targets have been added or removed, then DIFF will allow to classify the change. Examples of such scenarios include several environmental changes such as deforestation (trees are removed), coastal erosion (the coast line has been replaced by water), and some agricultural crops (ground is substituted by small stems, larger plants, and ground again). This is also useful for artificial targets (a vehicle appearing or disappearing) and urban (building built or demolished). It is important to note that this model also fits situations when the "visible" change is not clear. For example, in agriculture, when the plants get dry, the ground underneath get visible. Or when different moisture levels change the amount of surface scattering.
2) *Additive constrained:* If we have *a priori* knowledge that a unique partial target has been added or removed then ParDIFF will allow to classify the change in a better way. Examples include deforestation of sparse forests (we need to see the ground when the forest is present) and changes on crops. Practically, we still need to have portions of both partial targets $[C_1]$ and $[C_2]$ in both images to avoid convergence of the change matrix into the two original partial targets $[C_1]$ and $[C_2]$. Using a modification of ParDIFF that applies two optimizations instead of one, we can provide results similar to the unconstrained additive model, however it still suffers from convergence of the change matrix.
3) *Multiplicative:* This decomposition fits when the same target is modified and the parts that appear as being added or removed do not constitute proper interpretable polarimetric targets. For instance, if we rotate a car of 20°, it is very likely to show different SMs, but it is hard to argue that these are added to the scene, since the real point target producing them is the same. In this case, a multiplicative model is more appropriate, although retrieving information from the transformation may still be not straightforward and it may need the development of physical models.

Concerning detection capability, we expect the larger dynamic range of the RATIO to produce better point detection results. For instance, if we consider the signal to clutter ratio (SCR) as an indicator of separability, then RATIO optimizes this and therefore provides the highest SCR achievable. However, SCR is not the only parameter that impact detection performance and the standard deviation of target and clutter are also very important.

In the following, we are performing Monte Carlo simulations and use real data to test these theoretical considerations.

IV. Test With Simulated Data

A. Monte Carlo Simulations

In this simulation, we assume fully developed speckle with scattering vectors that can be modeled using 3-D zero-mean complex Gaussian. This assumption makes interpretation of the results easier, but interestingly it can also cover classification results when the texture does not change the structure of the covariance matrix.

The generation of the random variables is mostly following the processing steps proposed in [15].

1) A Monte Carlo method is used to generate N realizations of scattering vectors drawn by a 3-D Complex Circular Gaussian distribution. This is performed twice (one for each acquisition). Therefore, we generate two sets of "white random vectors": $\underline{k}_1^w(i)$ and $\underline{k}_2^w(i)$, where $i = 1, \ldots, N$.
2) The white random vectors are colored using two asymptotic covariance matrices $[\dot{C}_1]$ and $[\dot{C}_2]$ (representing the partial targets observed in the two acquisitions): $[\dot{C}_1]^{(1/2)}\underline{k}_1^w(i) = \underline{k}_1(i)$ and $[\dot{C}_2]^{(1/2)}\underline{k}_2^w(i) = \underline{k}_2(i)$.
3) The simulated covariance matrices are calculated by averaging N realizations of the outer product of simulated vectors: $[C_1] = \langle \underline{k}_1 \underline{k}_1^{*T} \rangle_N$ and $[C_2] = \langle \underline{k}_2 \underline{k}_2^{*T} \rangle_N$, where $\langle . \rangle_N$ is the finite average of N realizations. Since the simulation provides random variables that are close to be independent, the value of equivalent number of look (ENL) can be approximated by N.
4) Points 1–3 are repeated K times to provide more samples for evaluating statistics (e.g., they represent K experiments). In other words, a set of K covariance matrices is produced: $[C_1(k)]$ and $[C_2(k)]$, with $k = 1, \ldots, K$. Each of these realizations is slightly different due to speckle.
5) For each pair of the K experiments (e.g., $[C_1(k)]$ and $[C_2(k)]$), we evaluate the diagonalizations: $([C_2(k)] - [C_1(k)])\underline{\omega} = \lambda\underline{\omega}$ and $[C_1(k)]^{-1}[C_2(k)]\underline{\omega} = \lambda\underline{\omega}$.

To summarize, K samples are generated averaging N realizations and for each sample the change optimizations are applied. We, therefore, have two K sets of eigenvalues

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

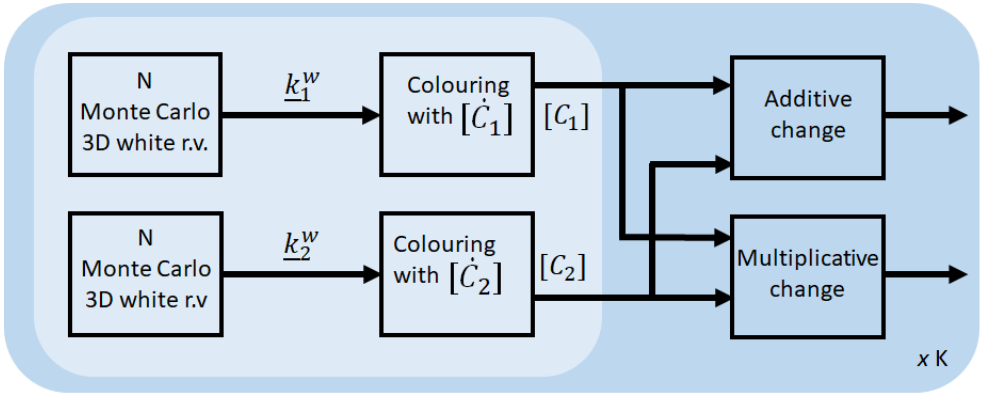


Fig. 1. Block diagram for the Monte Carlo simulations. rv: random variables. (Bottom right of shaded areas) $\times K$ indicates the number of times the operation is repeated.

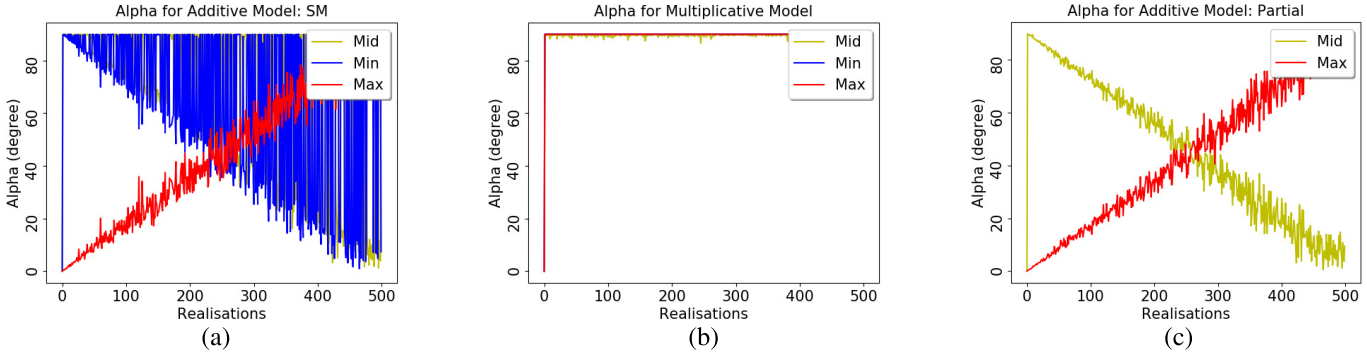


Fig. 2. Adding one target. Simulation of retrieved α values when (a) DIFF, (b) RATIO, and (c) ParDIFF optimizations. The dominant α of the target that has been added is swept from 0° to 90°. Entropy of targets = 0.1.

and eigenvectors, one set for each change detector. A block diagram of the simulation procedure is presented in Fig. 1.

B. Classifying Changes: Eigenvector Analysis

In this section, we test the capability of the three optimizations to classify a polarimetric change. In all the following simulations, we use an ENL equal to 50. This may correspond to an averaging window of around 11×11 for most of the current sensors. Since the purity of polarization of the partial target is very important, we perform simulations where we modify the entropy of the considered targets. The α values provided in the following are referred to the dominant component of the partial target using Pauli basis. We use this because all the optimizations focus on the dominant changes (or the first two dominant in case of ParDIFF) and α allows a clear and synthetic representation of the target characteristics.

In this section, we will be showing figures where the horizontal axis is the Realization (or repeat of the experiment), while the vertical axis is the dominant α angle that has been retrieved by the algorithm. The captions give information about the actual α values that are simulated in each of the plots.

1) Adding One Target: In the first test, the second partial target is obtained by adding 1 partial target to the first one. This follow the hypothesis H_1:$C_2 = C_1 + C_a$, where C_a is a unique partial target. This physical change meets the assumptions for both DIFF and ParDIFF. We want to assess the retrieval of the dominant SM of C_a.

Polarized targets: Fig. 2 shows a simulation where both targets C_1 and C_a are polarized (almost single targets) with an entropy of 0.1. The dominant α of the first target C_1 is zero (a surface) and the dominant α of C_a is swept from 0° to 90° (from surface to dihedral). A good retrieval will therefore estimate an α that goes from 0 to 90 along a line throughout the different realizations. The maximum eigenvector will estimate the dominant α of C_a and C_p, since this is the SM that has been maximally added to the scene. In the plot, this is shown as a red line. We can observe that DIFF and ParDIFF are able to identify the polarimetric characteristic of the added target, where the red line follows a ramp in the plot. On the other hand, the RATIO is not able to retrieve the correct α. Looking at the estimation errors, it seems both difference optimizations do not show any evident pattern and provide decent estimations with errors that are overall smaller for the ParDIFF with an RMS of 3.9° compared to 4.7 for DIFF. Note that ParDIFF minimum eigenvector is not shown because the matrix C_p is rank-deficient. Additionally, the DIFF minimum eigenvector (the target removed) is unstable (blue line), because nothing has been "removed" and the third eigenvalue is practically zero.

Partially polarized target: Fig. 3 shows a more challenging scenario when the entropy of $[C_1]$ and $[C_a]$ is 0.5. Note that in this simulation we are not adding a single target, but a proper partial target (formed by three single targets). DIFF and ParDIFF can only evaluate a portion of the total number of SM as discussed previously. This is why we are showing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

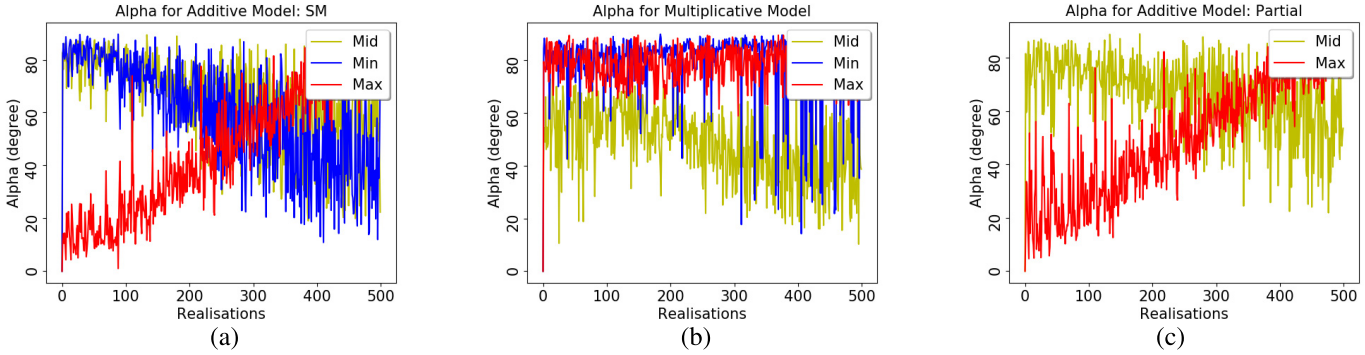


Fig. 3. Adding one target. Simulation of retrieved α values when (a) DIFF, (b) RATIO, and (c) ParDIFF optimizations. The dominant α of the target that has been added is swept from 0° to 90°. Entropy of targets = 0.5.

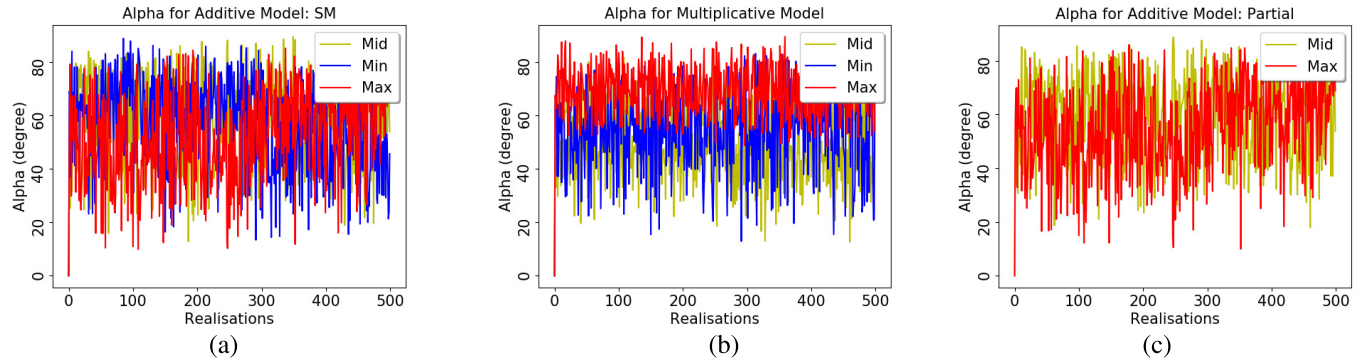


Fig. 4. Adding one target. Simulation of retrieved α values when (a) DIFF, (b) RATIO, and (c) ParDIFF optimizations. The dominant α of the target that has been added is swept from 0° to 90°. Entropy of targets = 0.99.

and concentrating only on the dominant change. As for the previous case, DIFF and ParDIFF can retrieve the added α, while RATIO provides completely unrelated values. Looking at the RMS error this has grown, where now DIFF can obtain a better result with an average 9.2° of error, while ParDIFF has 12.8°.

Depolarized target: The final test in Fig. 4 shows the same simulation when the entropy of C_1 and C_a is increased to 0.99. There is very little polarimetric information left and therefore retrieving any SM is very challenging. We can conclude that almost completely depolarized targets do not allow to retrieve α changes when the equivalent number of looks is 50. Things improve when averaging much more (e.g., $N = 200$), but these simulations are excluded for the sake of brevity. The RMS errors are now 27.8° for DIFF and 28.5° for ParDIFF.

The preliminary analysis shows that if the physical change is the addition of a unique partial target, DIFF and ParDIFF are able to retrieve the dominant SM that has been added to the scene, provided the entropy is not too high.

2) Adding and Removing Targets: In this more challenging scenario, we want to test if we can retrieve changes to two (and not one) partial targets. Specifically, we are adding a partial target AND removing another partial target. Note that this is the most general additive change and this assumption is not met by ParDIFF, which is only modeling the change to a unique partial target.

Please note in this experiment we are adding AND subtracting, considering both time directions (e.g., a dense forest, which becomes bare ground, the forest is subtracted and the ground is added). If we add (subtract) more partial targets in the same time direction, the superposition of effects will make it appears as a unique final partial target. That is H_1:$[C_2] = [C_1] + [C_{a1}] + [C_{a2}] \leftrightarrow [C_2] = [C_1] + [C_a]$, where $[C_a] = [C_{a1}] + [C_{a2}]$. In this case, ParDIFF will meet its assumptions and clearly, the only way to unpack the partial targets of origin $[C_{a1}]$ and $[C_{a2}]$ is by using scattering models.

The alternative hypothesis is now equal to H_1:$[C_2] = [C_1] + [C_a] - [C_s]$. The targets we want to retrieve now are two, the added $[C_a]$ and the subtracted $[C_s]$, which we will analyze using the dominant α parameters for the maximum and minimum eigenvectors, respectively.

In the first test, we select two orthogonal targets, where the dominant α of C_1 is zero, the α of C_a is 90° (a dihedral) and the α of C_s is 0. In this way, we have that $C_s = C_1$ and $C_a = C_2$. In practice, we are substituting a surface with a dihedral, both of equal power.

Polarized targets: Fig. 5 shows a plot where the entropy of all the targets is 0.1 (they are almost single targets). Interestingly, DIFF can retrieve both the targets added $\alpha = 90$ (red line) and removed $\alpha = 0$ (blue line). The RATIO cannot retrieve the α for the minimum, but the maximum is correct. The ParDIFF presents a very peculiar behavior where the maximum eigenvector jumps randomly between the target that has been added and the one that has been removed. The algorithm has not enough information to evaluate if a target has been added or if it has been removed and selects r_m or r_p based on consideration on total power [19], which in this case is simulated as being in average the same. However, if we fix

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

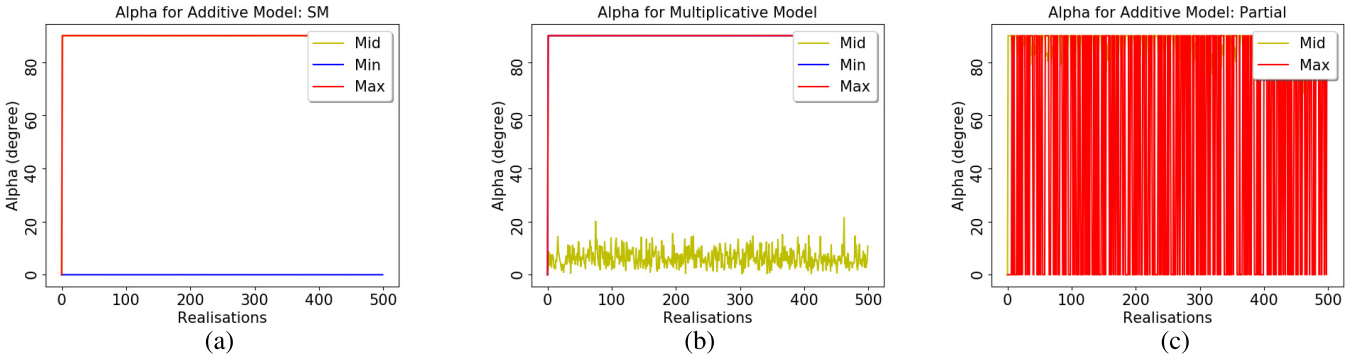


Fig. 5. Adding and removing targets. Simulation of retrieved α values when (a) DIFF, (b) RATIO, and (c) ParDIFF optimizations. The added α is 90°, and the removed α is 0°. Entropy of target equal to 0.1.

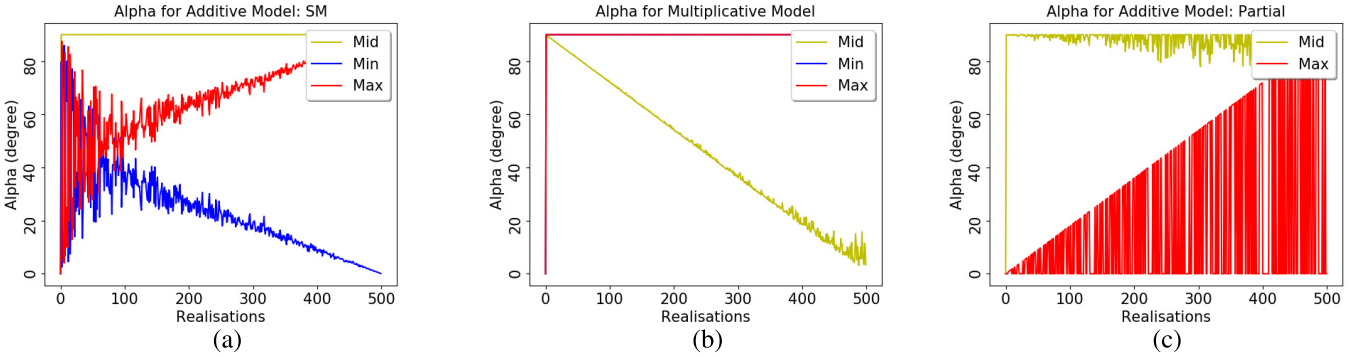


Fig. 6. Adding and removing targets. Simulation of retrieved α values when (a) DIFF, (b) RATIO, and (c) ParDIFF optimizations. The added α is swept between 0° and 90°, the removed α is 0°. Entropy of the target equal to 0.1.

the time direction, then the estimations become comparable to DIFF with RMS around 0.005°. The plots with ParDiff applied fixing the time direction are not shown for brevity.

Partially polarized targets: The same test was done when the entropy of all the targets is 0.5. For the sake of brevity, the figure is not shown, but the plots of α are very similar (just with an increased spreading). Again, DIFF retrieves both the dominant added target $\alpha = 90$ and the removed one $\alpha = 0$. The quality of the retrieval is still very good with estimation error of 2.8° (for DIFF). This shows that DIFF is able to cope with situations when one target is added and one removed. Again ParDIFF suffers from the same ambiguity where the algorithm does not know which direction to select and only detects "half" of the change. By fixing the time direction and running the algorithm twice, we can retrieve results just as good as the DIFF.

Depolarized targets: The figure when the entropy of all the targets is 0.99 is not shown for sake of brevity. Not surprisingly, the retrievals are very poor. However, it is still possible to see some pattern in the DIFF retrieval where we have a mean error of 24.1°. Interestingly, these results are much better than the one obtained when only one target was added. The reason for this is possibly that the change now is more marked than before and even a tiny amount of polarimetric information is magnified and more visible.

3) Adding and Removing Targets: Non Orthogonal: In the final test, we consider non-orthogonal targets. This means that the subtraction is not removing completely C_1 and the addition is not adding completely C_2. In particular, we have a surface which becomes a dipole. The dominant α for C_1 is 0, while the dominant α for C_2 is 45°. Clearly, mathematically there is no unambiguous combination of C_a and C_s that could have produced that change. We could consider an infinite number of different combinations that could have led to such result. What the optimization is able to tell us is the minimal addition C_a and subtraction C_s that would have transformed C_1 into C_2. This by definition leads to dominant SMs in C_a and C_s that do not have projections on each other (i.e., they are orthogonal).

If in nature, the change is obtained in a different way, we simply do not have enough information in the SAR data alone to disentangle this. However, we can always consider a scattering model that is able to start from the minimal changes and project these onto some target basis that we believe are responsible for the physical scattering. This can be done, for instance, in agriculture [30].

Polarized targets: Fig. 6 shows the simulation using non-orthogonal targets with entropy equal to 0.1. The α of the added target is swept between 0° and 90°, while the removed target has always an alpha equal to 0. We observe that the maximum and minimum of DIFF are not equal to the targets before and after. They are equal to the minimal change that would lead the first target to become the second. These two SMs will modify the surface and make it into the second target. Interestingly, if we look at the envelop of the red signal (top and bottom of red area), ParDIFF seems to be able to cope with this estimation, provided we fix the direction of time. This is because the operation that is making $[C_s]$ positive semi-definite is recognizing the assumptions are not met and estimating values of r_m and r_p very close to zero (around 10^{-7} in this experiment). This mean that $[C_s] \approx [C_1]$ or

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

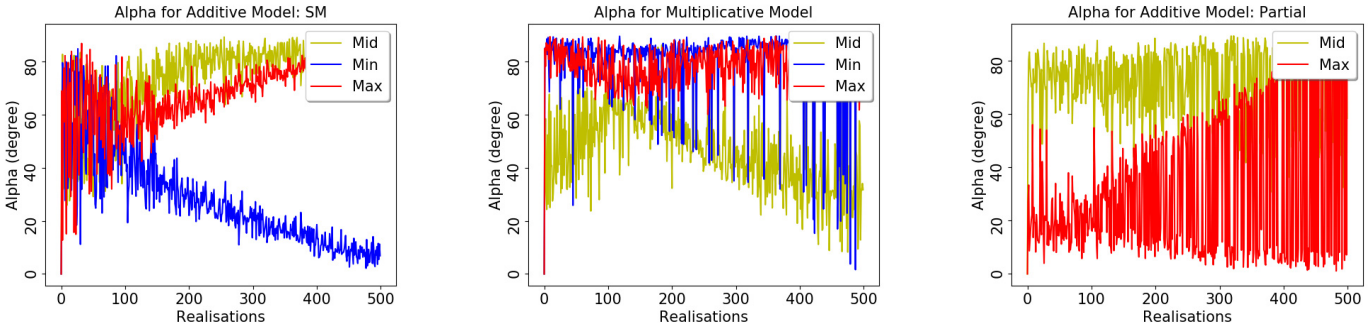


Fig. 7. Adding and removing targets. Simulation of retrieved α values when (a) DIFF, (b) RATIO, and (c) ParDIFF optimizations. The added α is swept between 0° and 90°, and the removed α is 0°. Entropy of the target equal to 0.5.

$[C_s] \approx [C_2]$. The change matrix converged to the partial targets in the two scenes. Clearly, the same result could have been obtained doing a diagonalization of $[C_1]$ or $[C_2]$ in the first place, but ParDIFF has the advantage to test the validity of adding a unique target and converges into the analysis of the separate images only when its own assumptions are not met.

Partially polarized targets: Fig. 7 shows the simulation using non-orthogonal targets with entropy equal to 0.5. As for the previous case, DIFF provides the α of minimum and maximum changes, while ParDIFF is able to recognize the model for a sole partial target is not true and returns the eigenvectors of the two matrices separately. Now the values of r_m and r_p are around 0.15. The differences between the estimation errors of DIFF and ParDIFF (provided we apply ParDIFF twice) start getting much smaller, because the change matrix needs to converge less toward the original targets.

Depolarized targets: similar to previous cases, none of the algorithms, can estimate the SMs and the results between DIFF and ParDIFF are very similar. The plots are not showed here for the sake of brevity.

4) Summary of Classification: From the simulation, we could produce the following conclusions.

1) *Additive models:* adding OR removing.
   a) DIFF can retrieve the two dominant SMs that have been added and/or removed.
   b) ParDIFF can retrieve one dominant SM either added or removed. If repeated twice disabling the automatic selection of the time direction, than it can retrieve the two dominant SMs.
2) *Additive models:* adding AND removing.
   a) DIFF retrieves the minimal changes to SM to go from $[C_1]$ to $[C_2]$. This information is complementary from knowing $[C_1]$ and $[C_2]$.
   b) ParDIFF recognizes its assumptions are not met and retrieves the two partial targets of origin or end (due to the change matrix converging toward the original partial targets).
3) RATIO produces eigenvectors that cannot be interpreted as SM.
4) When the entropy approaches 1, all the algorithms fail.

C. Detection: Eigenvalue Analysis

In this second simulation, we want to test the detection capability of the three change optimizations. To do this, we will produce receiver-operating characteristic curves for different situations and SCR values. These are plots showing Probability of Detection P_d against Probability of False Alarm P_f. The probability are calculated simulating realizations of targets (H_1) and clutter (H_0).

1) Adding One Target: The SCR is calculated considering the trace of C_a as our signal and the trace of C_1 as our clutter. This is because we want to detect an added target, embedded in the target of the first acquisition. For instance, a SCR = 0.5 means that the trace of C_a is half the one of C_1. We simulated 10 000 instances of the first target and 10 000 instances of the second target, to calculate P_d and P_f.

In these simulations as benchmark, we considered the Wishart change detector based on the likelihood ratio [18]. This was developed by Conradsen *et al.* [18] and it is sometime referred to as the Conradsen change detector. In the following figures, we will show ROC curves, where we express the probability of false alarm P_f with a base 10 logarithmic scale. In each of these plots, the color of the line identifies the algorithm, while the style (solid, dashed, dotted) identifies the value of SCR.

Polarized targets: Fig. 8 shows the ROC curves when $[C_1]$ and $[C_a]$ have an entropy equal to 0.1 (note $[C_2] = [C_1] + [C_a]$). The solid line is for SCR = 0.5, dashed line is for SCR = 0.1, and the dotted line (appearing as thicker in the plot) is for SCR = 0.05. The black line (the lowest performance possible) represents the random assignment (flipping a coin and deciding if it is target or clutter). Note that the scale for P_f is in $\log_{10}$.

Interestingly, the performances of RATIO and Wishart are top of the scale for SCR = 0.5 and for SCR = 0.1. They provide perfect detection in our simulations (i.e., $P_d = 1$ and $P_f < 10^{-4}$) and they cannot be seen because they overlap on each other at the top of the ROC plot (only visible a blue line, the green line is underneath). Note that with 10 000 simulations, we can only estimate $P_f < 10^{-4}$. Since this is a good limit for change detection, we did not simulate more samples which would have led to a slowdown of the computation. The DIFF is providing lower detection performance, which we may have expected considering the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

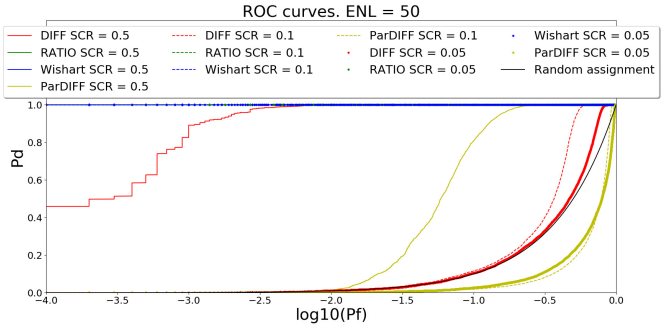


Fig. 8. ROC curves, one target added. Simulation for different SCR when the first target is a surface and the target added has a dominant $\alpha = 90°$. Entropy of targets equal to 0.1.

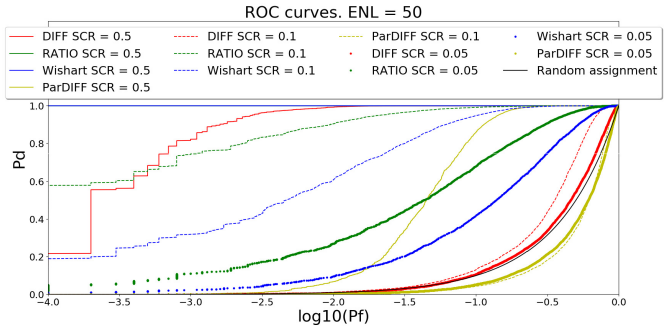


Fig. 9. ROC curves, one target added. Simulation for different SCR when the first target is a surface and the target added has a dominant $\alpha = 90°$. Entropy of targets equal to 0.5.

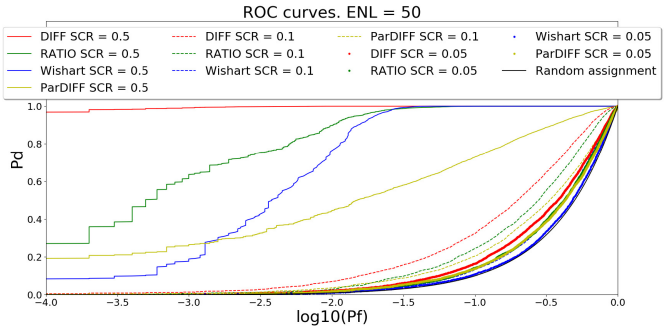


Fig. 10. ROC curves, one target added. Simulation for different SCR when the first target is a surface and the target added has a dominant $\alpha = 90°$. Entropy of targets equal to 0.99.

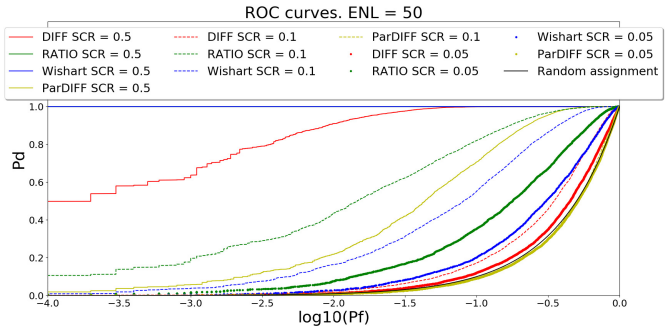


Fig. 11. ROC curves, one target added. Simulation for different SCR when the first target is a surface and the target added has a dominant $\alpha = 45°$. Entropy of targets equal to 0.5.

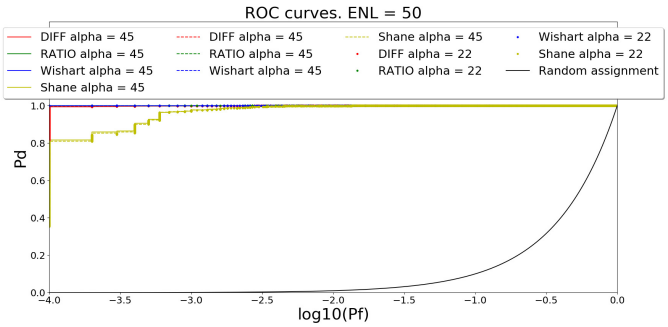


Fig. 12. ROC curves, targets added and removed. Simulation for different α of the second target (90°, 45°, and 22°). Entropy of targets equal to 0.5.

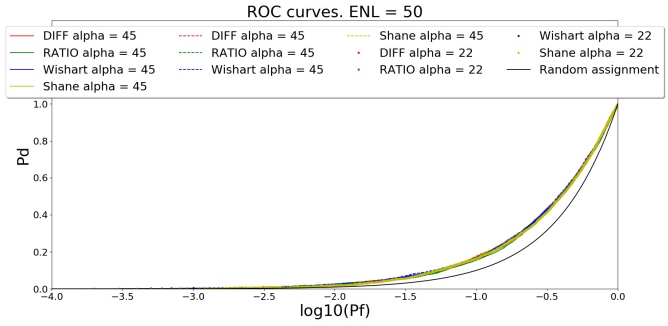


Fig. 13. ROC curves, one target added. Simulation for different SCR when the first target is a surface and the α added is 45°. Entropy of the target is 0.99.

ratios have larger dynamics. Finally, ParDIFF is providing the worst performance. This can be justified by the fact that to produce a positive semi-definite change matrix, we reduce the difference (r_s and r_m are generally smaller than 1). The power of the test is therefore reduced while this is not affecting false alarms.

Also note that when the SCR is very low (the added target is 5% of the initial target), the detection performance is very low for all the detectors resembling the random assignment.

Partially polarized targets: Fig. 9 shows the ROC curves when C_1 and C_a have an entropy equal to 0.5. Again the RATIO and Wishart provide the best results and performance degrades when the SCR reduces (green and blue lines overlap again). For SCR $= 0.5$, we have that RATIO and Wishart obtain the perfect performance, with DIFF again inferior in detection performance. For SCR $= 0.1$, RATIO and Wishart have a degraded performance and we can clearly see that RATIO performs better than Wishart. For SCR $= 0.05$, no algorithm is able to provide decent performance.

Depolarized targets: Fig. 10 shows the ROC curves when C_1 and C_a have an entropy equal to 0.99. A surprising thing happens here when the SCR is high enough (SCR $=$ 0.5). Once polarimetry becomes less important because the depolarization is really high, DIFF performs much better than other algorithms. When it comes to detection, DIFF sees more separability when the polarimetric information is less important and the discrimination is done purely on the total intensity of the partial targets. This is a result that we will see in real data as well. However, the good detection performance of DIFF can only be achieved when the SCR is large enough. If SCR $= 0.1$, no detector can provide a decent performance.

To summarize these first results, when we observe single targets (e.g., vehicles, small buildings), then we will have larger discrimination if we use RATIO or Wishart. On the other

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

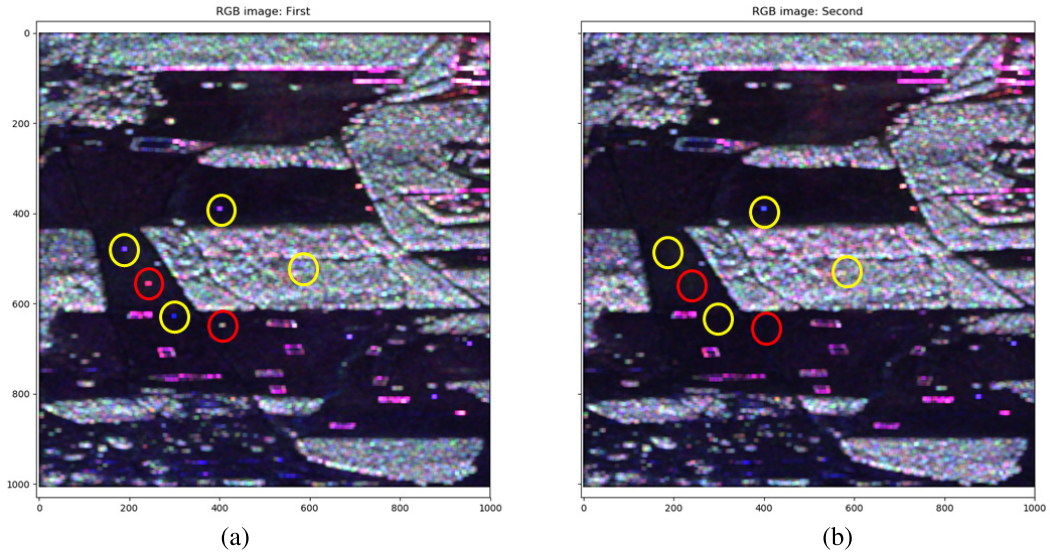


Fig. 14. RGB Pauli images of the SARTOM test site with four days of temporal baseline. (a) Master acquisition. (b) Slave acquisition (four days after). Circles indicate targets that were removed/moved. Yellow is for CRs, while Red were jeeps (DLR E-SAR L-band SARTOM2006 Campaign).

hand, when dealing with high entropy systems, we would get better detection performance by using DIFF.

Finally, in Fig. 11, we can observe the ROC curves when we add a dominant dipole $\alpha = 45$. The entropy is 0.5. We can see that the performance reduces especially for RATIO and Wishart with lower SCR, but the ranking of the detectors remains the same.

2) Adding and Removing Targets: In this section, we want to simulate the general situation when we add and remove targets. In the following simulations, we always start with an initial target $[C_1]$ equal to a surface and modify this into a second target $[C_2]$ with a variable dominant α (choosing between 90°, 45°, and 22°).

Polarized targets: when the entropy is 0.1, we can obtain perfect detection for all the algorithms. The ROC curve is not shown for the sake of brevity.

Partially polarized targets: The situation is identical to Fig. 12 and not showed here for the sake of brevity. An entropy of 0.5 still provides perfect detection (i.e., $P_d = 1$ and $P_f < 10^{-4}$). Targets with a difference of 22° cannot be discriminated using 50 ENL.

Depolarized targets: If we simulate very depolarized targets with entropy 0.99 (see Fig. 13), we discover that all the detectors perform very bad. The problem is that there is very small polarimetric difference with entropy 0.99 and the SCR of the two targets is identical. Therefore, DIFF cannot separate the targets based on the overall power as it was doing before for depolarized targets with different Trace. To have separability when there is no polarimetric information, we need a difference in overall intensity.

V. Tests With Real Data

A. Data Presentation: SARTOM E-SAR (DLR)

The first dataset was acquired by the E-SAR DLR system and it is L-band quad-polarimetric. The data collection called SARTOM [31] was performed in 2006 and it was mainly aimed at target detection using tomography and polarimetry. Interestingly, several targets were deployed in open field and under vegetation. Additionally, some of the targets were moved during the acquisitions. Fig. 14 shows the RGB Pauli images of the test area for two separated acquisitions. The spatial baseline is zero (in average) and the temporal one is four days. Several targets among two trihedral corner reflectors (CRs) in open field and one CR in the forest were removed during the four days.

B. Change Models on SARTOM

The three optimizations were applied to the SARTOM data and Fig. 15 shows the images of the eigenvalues.

All the targets with validated changes can be easily detected in the maximum eigenvalues of the RATIO (some in the maximum and some in the minimum). This includes the CRs under canopy cover. Interestingly, using only the backscattering, the latter cannot be detected. Moreover, some further point targets have a relatively large RATIO. Note that the northernmost of the CRs was positioned on the ground without pedestal. This was removed at the end of the first acquisition and then replaced on the ground four days after. It was not used for calibration purposes, but it only functioned as a target for detection. We can still detect the change, because its different positioning made it look different in the two images (looking at the Pauli RGB images reveals that the color around the CR looks different in the two images).

Targets are also visible in the DIFF and ParDIFF eigenvalues; however, the forest clutter is very high and an automatic change detector may struggle separating the point targets change from the background. Note that the scale of these two is different and this explains most of the differences in colors.

Fig. 16 shows the retrieved α using the two optimizations. Before interpreting these results, it is important to keep in mind that α is a scale-invariant measurement and it will be retrieved regardless of the intensity of the eigenvalue. This makes α images look very noisy. In the following, we propose

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

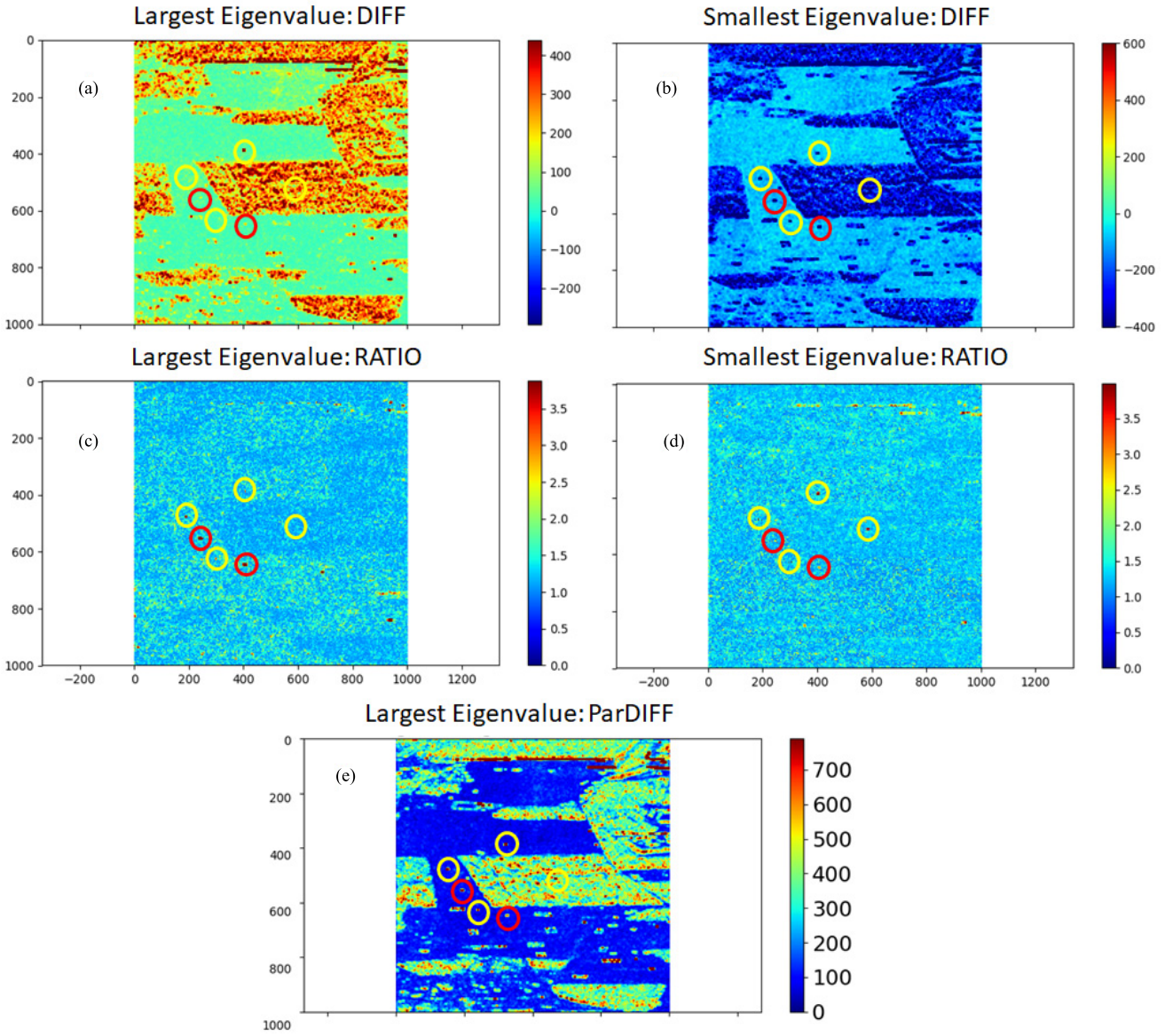


Fig. 15. Eigenvalues (SARTOM). (a) Largest Eigenvalue of DIFF. (b) Smallest Eigenvalue of DIFF. (c) Largest Eigenvalue of RATIO. (d) Smallest Eigenvalue of RATIO (if smaller then 1, the inverse is shown). (e) Largest Eigenvalue of ParDIFF. Circles indicate targets that were removed/moved. Yellow is for CRs, while Red were jeeps. Averaging: 9×9 boxcar.

a better way to visualize the α; however, it is interesting to have a look at α everywhere for testing purposes.

Looking at DIFF and ParDIFF, the images show a reduction of surface scattering on the fields. As expected, the α retrieved by the ratio does not seem to be interpretable as an SM.

Fig. 17 is built by using α as the color and the eigenvector as the saturation. A pixel will keep the color given by α, but appear very dark if the eigenvalue is low. Looking at the moved targets, we can observe how the removed SMs (smallest eigenvalue) resemble in colors the Pauli RBG of the first acquisition. The CRs appear mostly blue, although the northernmost one has a strange behavior when redeployed showing different colors. The CR in the forest is very hard to identify, due to the large clutter represented by forest changes. From the simulation, we saw that changes of depolarized targets seem to be enlarged by DIFF.

Finally, Fig. 18 presents the same color image where the RATIO optimization is performed. Interestingly, the detection of point targets is improved using the ratio, but the colors do not suggest any possibility to perform classification.

Fig. 19 presents the same color image where the ParDIFF model is performed. Interestingly, the RGB seems very similar to a mix of the largest and smallest RGBs of the DIFF. Simulations did show that both DIFF and ParDIFF are able to identify the changing SMs. Finally, Fig. 19 also presents a change detection image obtained using the Wishart change detector tested in the simulation section. Obviously, the Wishart detector will not be able to classify the change (it is black and white). Additionally, some of the targets, as the ones in the forest are almost invisible in the image, showing that in real data RATIO may have some advantage to the Wishart. This could be linked to the fact that simulations were assuming a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

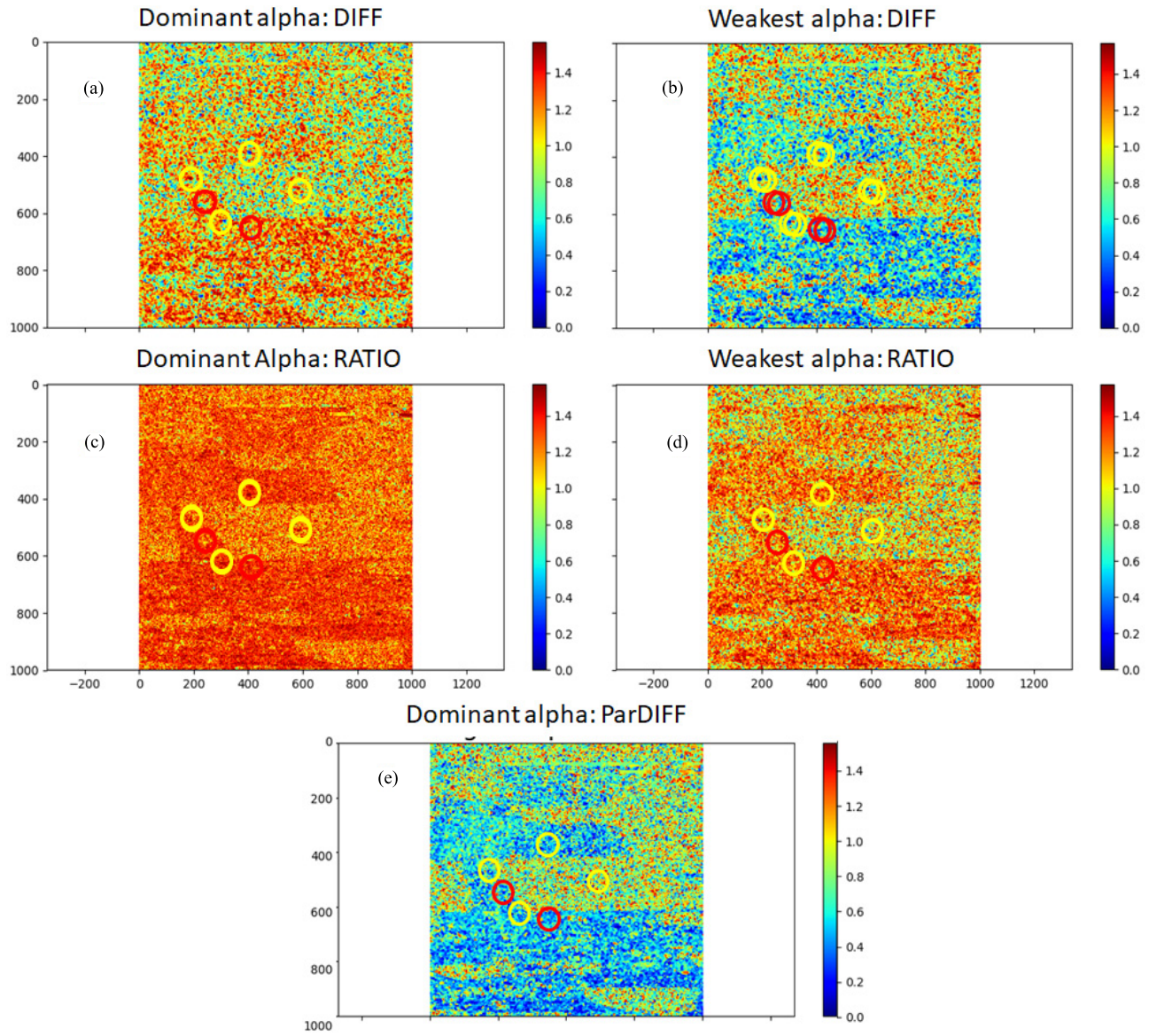


Fig. 16. α (SARTOM). (a) α for the largest eigenvector of DIFF. (b) α for lower eigenvector of DIFF. (c) α for the largest eigenvector of RATIO. (d) α for the lowest eigenvector of RATIO. (e) α for the largest eigenvector of ParDIFF. Averaging: 9×9 boxcar.

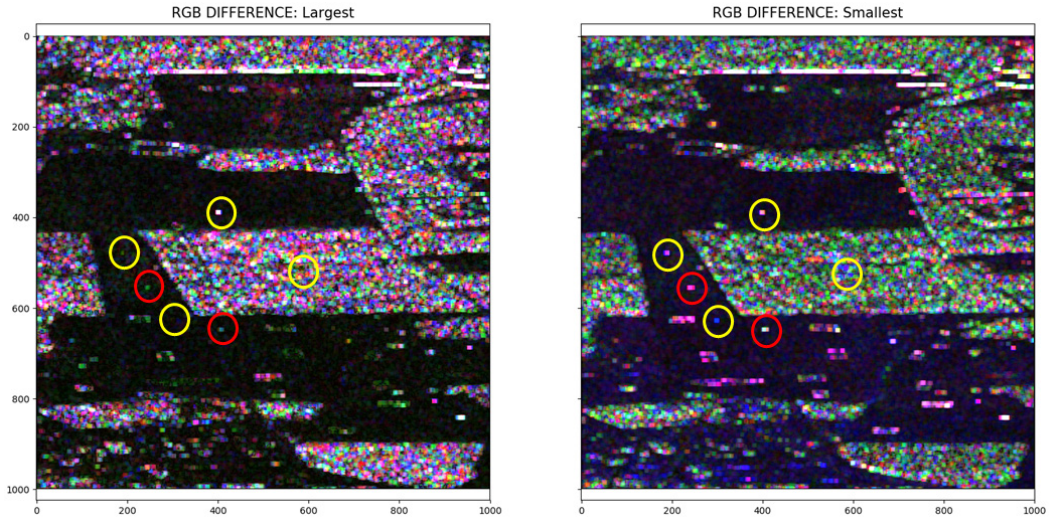


Fig. 17. DIFF optimization (SARTOM). (Left) Largest eigenvalue. (Right) Lower eigenvalue. Averaging: 9×9 boxcar.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

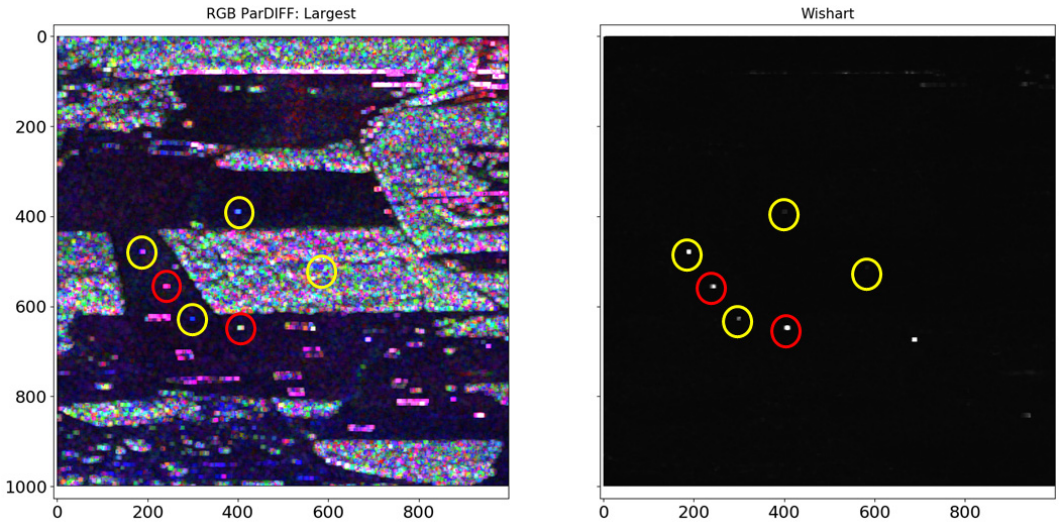


Fig. 18. Changed detection (SARTOM). (Left) ParDIFF optimization largest eigenvalue. (Right) Wishart. Averaging: 9×9 boxcar.

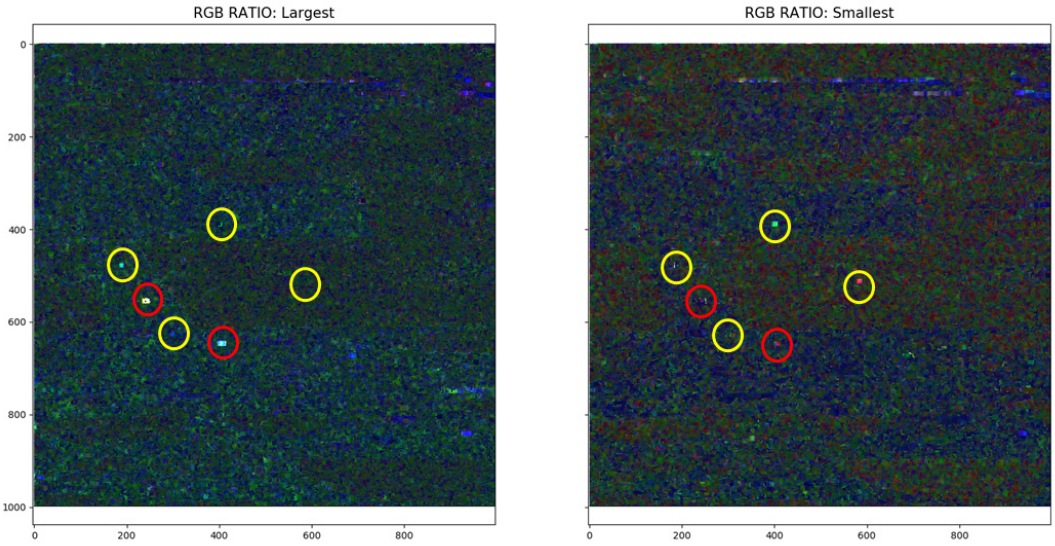


Fig. 19. RATIO optimization (SARTOM). (Left) Largest eigenvalue. (Right) Lower eigenvalue. Averaging: 9×9 boxcar.

Wishart distribution of the target and clutter, while real data may have a different distribution due to texture.

Finally, we want to investigate further the similarity of DIFF and ParDIFF on real data. To do so, Fig. 20 plots a 2-D density distribution of DIFF against ParDIFF estimations. If the density is on the bisect, then the two results are identical. The three plots represent the three elements of the Pauli decomposition that we used for the RGB. The estimations are in pretty good agreement, as indicated by the RMS and bias. ParDIFF tends to estimate more surface change than DIFF. From this comparison, we can conclude that indeed the two algorithms are looking at very similar SMs.

C. Data Presentation: ALOS-1 (JAXA)

The second dataset was acquired by the PALSAR instrument on board of the JAXA ALOS satellite. The images analyzed here were acquired in 2007 in the west of the U.K., over the Morecombe Bay.

The first image was acquired on April 1, while the second was on May 17, 2006. The area presents a very dynamic portion of the coast (with a salt marsh in the red rectangle) and a caravan park included in the red circle.

The Pauli RGB of the two images are presented in Fig. 21. It is possible to observe how the sea region is showing some changes due to different wind conditions. Additionally, a large area of agricultural fields near the coast shows an increased green color (indicating an increase in volume scattering). The caravan park also presents several point target changes.

1) Change Detection With ALOS Data: The resolution is 30 m $\times$ 10 m in slant range and azimuth. The coherency matrices were filtered with a 3×9 boxcar and multilooked to produce a pixel on the ground that is more squared. For the sake of brevity, we only present the final RGB images.

In Fig. 22, we present the α angle saturated by the eigenvalues for the DIFF optimization. Interestingly, DIFF is able to classify the changes to SMs, for instance, identifying an increase in surface scattering over the sea. Also DIFF identifies an increase of volume scattering over the agricultural areas.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

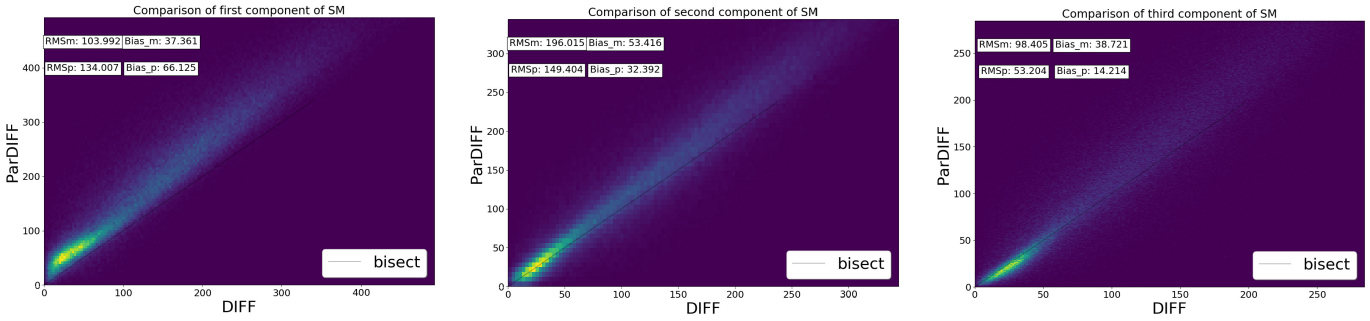


Fig. 20. Comparison (SARTOM). (Left) Largest eigenvalue. (Right) Lower eigenvalue. Averaging: 9×9 boxcar.

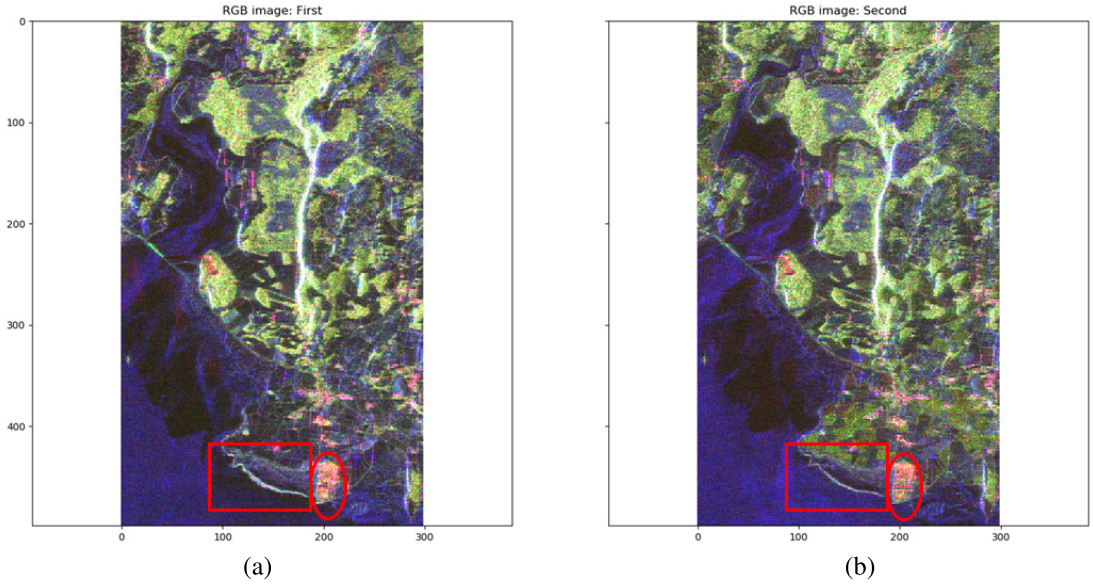


Fig. 21. RGB Pauli images of the JAXA test site with one and a half month of temporal baseline. (a) April 1, 2007. (b) May 17, 2007. (JAXA L-band.)

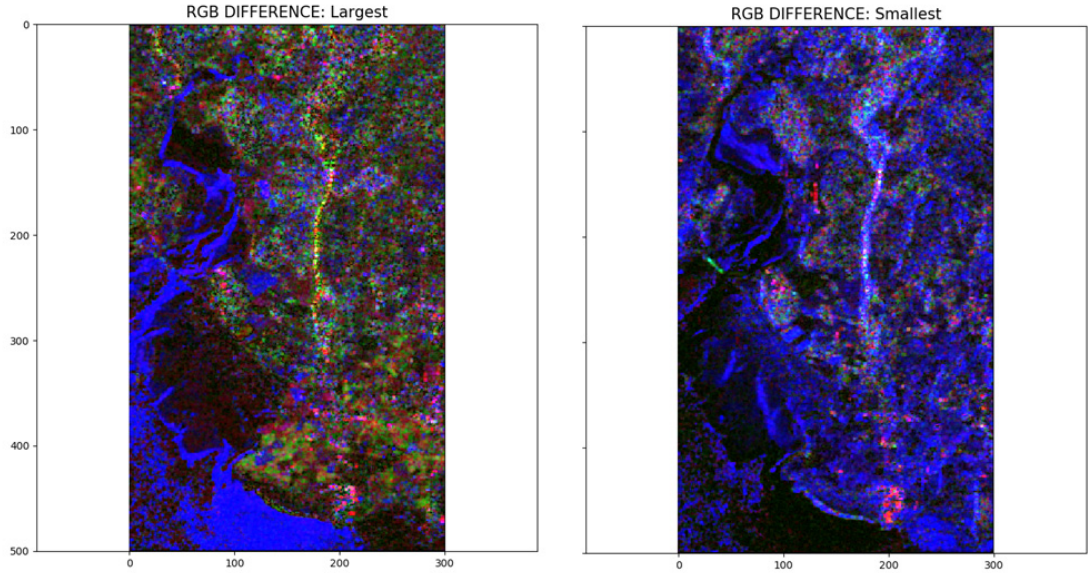


Fig. 22. DIFF optimization (JAXA). (Left) Largest eigenvalue. (Right) Lower eigenvalue. Averaging: 3×9 boxcar.

It monitors an increase and reduction of dihedral scattering over the caravan park (due to large moving vehicles).

Fig. 23 presents the same color image where the RATIO is performed. These images cannot be used to classify the changes. Additionally, they seem more appropriate for point target detection.

Fig. 24 presents the same color image where the ParDIFF is performed. Again the image seems to be a mix of the two

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

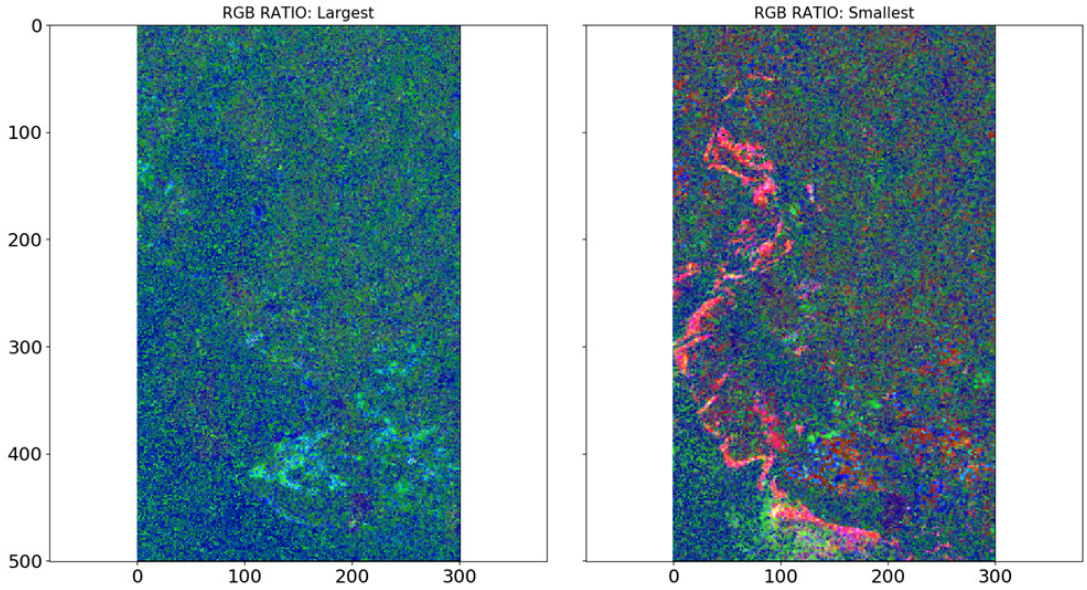


Fig. 23. Ratio optimization (JAXA). (Left) Largest eigenvalue. (Right) Lower eigenvalue. Averaging: 3×9 boxcar.

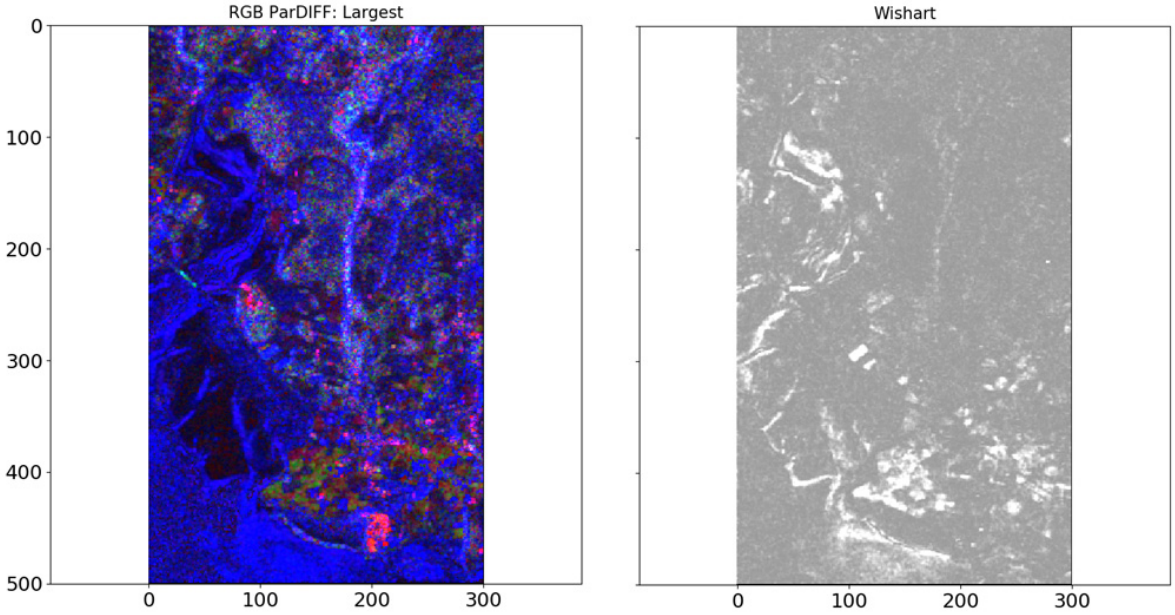


Fig. 24. ParDIFF optimization (JAXA). (Left) Averaging: 3×9 boxcar.

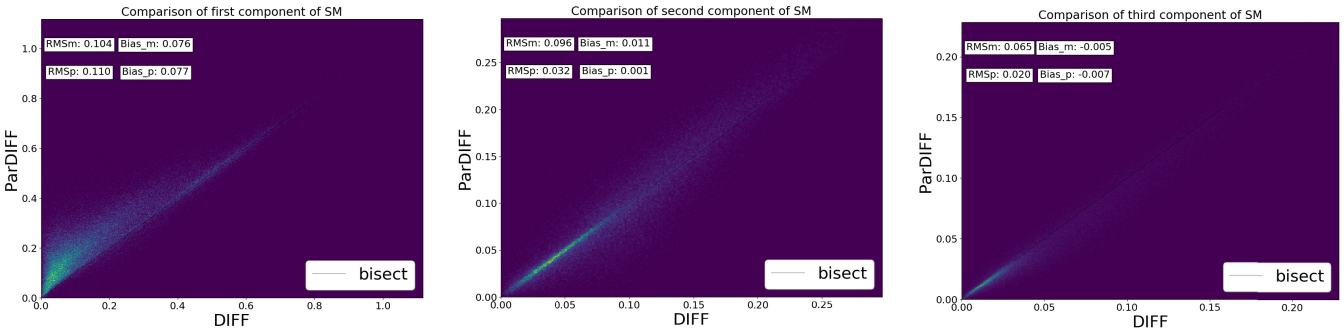


Fig. 25. Comparison (ALOS). (Left) Largest eigenvalue. (Right) Lower eigenvalue. Averaging: 9×9 boxcar.

DIFF images. The Wishart change detector is also presented and clearly it can detect but not classify the change.

To test again the similarity of outputs between DIFF and ParDIFF, Fig. 25 plots a density map of estimations. Again the three plots are the three elements of the Pauli decomposition. The RMS and bias show minimal differences of more than an order of magnitude smaller than the actual values. From this comparison, we can again conclude that the simulations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

were realistic and DIFF and ParDIFF produce similar results because they are observing the same underlying physical phenomena.

VI. Conclusion

In this work, a validation and comparison of three optimizations for change detection with PolSAR was carried out. One of these optimizations was also first published in this article. The algorithms are based on diagonalizations of change matrices and therefore they can be referred as spectral decompositions of the change matrix. These have the aim to separate the different components of signal representing a polarimetric change. The proposed decomposition is aimed at optimizing the DIFFERENCE (or DIFF) of covariance matrices $[C_a] = [C_2] - [C_1]$, while the second considers the RATIO of powers which lead to the change matrix $[C_m] = [C_1]^{-1}[C_2]$. The last was recently proposed [19] and aims at constraining the change matrix of DIFF to be positive semi-definite. This decomposition is here referred as ParDIFF.

The DIFF and ParDIFF decompositions can be used to provide physical interpretation to additive changes. This is when SMs are added or subtracted from the scene. The constrain applied to ParDIFF assumes that only one partial target is either added or removed, while DIFF does not need this assumption. Finally, RATIO could provide physical interpretation only when a transformation modifies the first partial target into the second one (e.g., a change in orientation).

It is important to notice that these are signal decompositions focused on changes and not targets. They therefore have different properties from ordinary Target Decompositions [11], [13].

The three decompositions were validated using Monte Carlo simulations. It was evident that DIFF can retrieve the dominant SMs that were added or removed, while RATIO was not providing this information. ParDIFF can also retrieve the two dominant SM changes, if applied twice to cover the two time directions ($1 \rightarrow 2$ and $2 \rightarrow 1$). On the other hand, RATIO and Conradsen detectors [18] provided better detection performance for polarized point targets.

Finally, the algorithms were tested on two different quad-polarimetric L-band datasets. The first was acquired during the E-SAR DLR SARTOM 2006 campaign and it is largely dedicated to point target detection. The second considers JAXA ALOS data acquired in 2006 over Morecombe Bay in U.K. This dataset was very beneficial to assess detection and classification of distributed targets. Interestingly, the eigenvectors result from DIFF and ParDIFF are very similar showing that both algorithms are able to identify the dominant changes in the scene.

1) *DIFF:* When the additive hypothesis is true, DIFF can represent the dominant SMs both added and removed. In case that there are multiple SMs added or removed, the decomposition is focusing on the unique minimal change that can lead from the first to the second target. This is ensured by the property of orthogonality of the eigenvectors. Since it focuses on the minimal change, its information is complementary to applying a spectral decomposition of $[C_1]$ and $[C_2]$ separately and looking at differences in parameters.
2) *ParDIFF:* When the additive hypothesis is true and only one partial target is either added or removed, ParDIFF is equivalent to a target decomposition because the change matrix will represent a physically feasible partial target. However, if there are multiple partial targets added or removed, the algorithm will not be able to follow both changes and it needs to be applied twice in the two time directions. Interestingly, when the assumptions are not met, the algorithm is able to deal with the situation by converging toward the covariance matrices of destination (first or second depending on time direction). Although the result is equivalent to apply a diagonalization of the original covariance matrix, ParDIFF is able to automatically recognize when this is needed. As a note of caution, applying the diagonalization in both directions may result in one of the matrices not being positive semi-definite.
3) *RATIO:* provides eigenvectors that cannot be interpreted as SMs, but directions in which we stretch the ellipsoid of the first partial target to make it into the second partial target. They are therefore very different from Target Decompositions and provide complementary information.

In terms of computational cost, DIFF presents the quickest algorithm with one diagonalization of a symmetric matrix per pixel. RATIO performs one diagonalization of a Normal, but asymmetric matrix and ParDIFF requires two diagonalizations if applied with the automatic selection of time direction (one symmetric and one symmetric) or three diagonalizations if applied twice.

To summarize, the DIFF and ParDIFF decompositions are suited to classify the type of change (if the additive hypothesis is fulfilled), while RATIO is better in detecting polarized point targets.

As a future work, we want to explore more optimizations of different change matrices. We want also to investigate how to combine physical models to interpret the changes (e.g., for agricultural purposes) and we want to extend these decompositions to time series of PolSAR data.

Acknowledgment

The SARTOM2006 were acquired by the E-SAR airborne system of the German Aerospace Center (DLR), Wessling, Germany. The authors would like to thank the DLR Team that participated to the SARTOM campaign in 2006, in particular Ralf Horn and Rolf Scheiber. The ALOS data were provided by JAXA under Project 1151. ALOS Product-JAXA 2006, all rights reserved.

References

[1] M. Preiss and N. J. S. Stacy, *Coherent Change Detection: Theoretical Description and Experimental Results*. Fairbairn, CBR, Australia: Defence Science and Technology Organisation, 2006.
[2] E. J. M. Rignot and J. J. van Zyl, "Change detection techniques for ERS-1 SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 31, no. 4, pp. 896–906, Jul. 1993.
[3] R. J. Dekker, "SAR change detection techniques and applications," in *Proc. 25th EARSeL Symp. Global Develop. Environ. Earth Observ. Space*, Porto, Portugal, 2005, pp. 63–69.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.

[4] R. J. Radke, S. Andra, O. Al-Kofahi, and B. Roysam, "Image change detection algorithms: A systematic survey," *IEEE Trans. Image Process.*, vol. 14, no. 3, pp. 294–307, Mar. 2005.
[5] J.-F. Mas, "Monitoring land-cover changes: A comparison of change detection techniques," *Int. J. Remote Sens.*, vol. 20, no. 1, pp. 139–152, 1999.
[6] L. M. Novak, "Coherent change detection for multi-polarization SAR," in *Proc. Conf. Rec. 39th Asilomar Conf. Signals, Syst. Comput.*, 2005, pp. 568–573.
[7] A. Marino, S. R. Cloude, and J. M. López-Sanchez, "A new polarimetric change detector in radar imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2986–3000, May 2013.
[8] A. A. Nielsen, K. Conradsen, and H. Skriver, "Change detection in full and dual polarization, single- and multifrequency SAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 8, pp. 4041–4048, Aug. 2015.
[9] A. A. Nielsen, H. Skriver, and K. Conradsen, "The Loewner order and direction of detected change in Sentinel-1 and Radarsat-2 data," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 242–246, Feb. 2020.
[10] A. A. Nielsen, "Fast matrix based computation of eigenvalues and the Loewner order in PolSAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 10, pp. 1727–1731, Oct. 2020.
[11] Y. Yamaguchi, T. Moriyama, M. Ishido, and H. Yamada, "Four-component scattering model for polarimetric SAR image decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 8, pp. 1699–1706, Aug. 2005.
[12] A. Freeman and S. L. Durden, "A three component scattering model for polarimetricsar data," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 3, pp. 963–973, 1998.
[13] S. R. Cloude and E. Pottier, "A review of target decomposition theorems in radar polarimetry," *IEEE Trans. Geosci. Remote Sens.*, vol. 34, no. 2, pp. 498–518, Mar. 1996.
[14] S. R. Cloude, *Polarisation: Applications in Remote Sensing*. Oxford, U.K.: Oxford Univ. Press, 2009.
[15] J. S. Lee and E. Pottier, *Polarimetric Radar Imaging: From Basics to Applications*. Boca Raton, FL, USA: CRC Press, 2009.
[16] A. Marino and A. Alonso-González, "Optimisations for different change models with polarimetric SAR," in *Proc. EUSAR Conf.*, 2018, pp. 1–6.
[17] A. Swartz, H. Yueh, J. Kong, L. Novak, and R. Shin, "Optimal polarizations for archiving maximum contrast in radar images," *J. Geophys. Res.*, vol. 93, pp. 252–260, Dec. 1988.
[18] K. Conradsen, A. A. Nielsen, J. Schou, and H. Skriver, "A test statistic in the complex Wishart distribution and its application to change detection in polarimetric SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 1, pp. 4–19, Jan. 2003.
[19] S. R. Cloude, "A physical approach to POLSAR time series change analysis," *IEEE Geosci. Remote Sens. Lett.*, early access, Sep. 14, 2020, doi: 10.1109/LGRS.2020.3019936.
[20] S. R. Cloude and E. Pottier, "An entropy based classification scheme for land applications of polarimetric SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 1, pp. 68–78, Jan. 1997.
[21] S. M. Kay, *Fundamentals of Statistical Signal Processing*. Upper Saddle River, NJ, USA: Prentice-Hall, 1993.
[22] A. Marino and I. Hajnsek, "A change detector based on an optimization with polarimetric SAR imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 8, pp. 4781–4798, Aug. 2014.
[23] A. Alonso-Gonzalez, C. Lopez-Martinez, and P. Salembier, "PolSAR time series processing with binary partition trees," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 6, pp. 3553–3567, Jun. 2014.
[24] A. Alonso-González, K. Papathanassiou, and I. Hajnsek, "Physical interpretation of the polarimetric SAR changes observed over agricultural time series," BioGeo SAR, Harwell, U.K., Tech. Rep. 24, 2015.
[25] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 1, pp. 234–244, Jan. 1993.
[26] R. O. Duda, P. E. Hart, and D. H. Stork, *Pattern Classification*. Hoboken, NJ, USA: Wiley, 2000.
[27] G. Strang, *Linear Algebra and its Applications*. Boston, MA, USA: Thomson Learning, 1988.
[28] A. Marino, "Comparison of signal models for change detection with polarimetric," in *Proc. ESA POLinSAR Workshop*, 2018, pp. 1–6.
[29] A. Marino and A. Alonso-Gonzalez, "An optimization of the difference of covariance matrices for PolSAR change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5315–5318.
[30] C. Silva, A. Marino, J. Lopez-Sanchez, and I. Cameron, "Monitoring agricultural fields using an optimisation of the difference of covariance matrices for PolSAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 6619–6622.
[31] R. Horn, M. Nannini, and M. Keller, "SARTOM airborne campaign 2006: Data acquisition report," DLR HR, Wessling, Germany, Tech. Rep. DLR-HR-SARTOM-TR-001, 2006.



Armando Marino (Member, IEEE) received the M.Sc. degree in telecommunication engineering from the Università di Napoli "Federico II," Naples, Italy, in 2006, and the Ph.D. degree in polarimetric synthetic aperture radar (SAR) interferometry from the School of Geosciences, University of Edinburgh, Edinburgh, U.K., in 2011.

In 2006, he joined the High Frequency and Radar Systems Department, German Aerospace Center, Wessling, Germany, where he developed his M.Sc. thesis. From March 2011 to October 2011, he was with the Institute of Computing Research, University of Alicante, Alicante, Spain. From December 2011 to May 2015, he was a Post-Doctoral Researcher and a Lecturer with the Institute of Environmental Engineering, ETH Zurich, Zurich, Switzerland. In June 2015, he was a Lecturer with the School of Engineering and Innovation, Open University, Milton Keynes, U.K. Since May 2018, he has been a Senior Lecturer (an Associate Professor) with the Faculty of Natural Sciences, University of Stirling, Stirling, U.K.



Matteo Nannini received the Laurea degree in telecommunication engineering from the Università degli studi di Firenze, Florence, Italy, in 2003, with a thesis done in collaboration with the Microwaves and Radar Institute (HR), German Aerospace Center (DLR), Wessling, Germany, and the Ph.D. degree from the Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany, in 2009, with a thesis on synthetic aperture radar (SAR) tomography.

Since 2004, he has been with DLR-HR. His main research interests include SAR processing, differential interferometry, tomography, and radar sounding, including mission design and performance analysis of future SAR missions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 04:29:28 UTC from IEEE Xplore. Restrictions apply.